P25

Abstract No. 24JSDP

Foundation-model perturbation analysis identifies T-cell dysfunction programs in small cell lung cancer

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Code, result tables, notebooks & drafts

24JSDP | Final | single-cell → spatial pathology | **COI:** The authors declare no conflict of interest. | HTAN MSK CELLxGENE · GSE263196 10x Visium · Geneformer-V2-104M

Take-home: model-derived dysfunction candidates in SCLC → **antigen-presentation program** → associated with T-cell abundance in independent tumour tissue

A · QUESTION, COHORT & METHOD

WHY THIS MATTERS

Small cell lung cancer (SCLC) is the most aggressive thoracic malignancy and responds poorly to immune checkpoint blockade despite a high tumour mutational burden. T-cell dysfunction in the tumour microenvironment is a candidate mediator, but its transcriptional basis is poorly characterised at single-cell resolution.

Gap. Conventional differential expression captures mean shifts and misses non-linear, context-dependent regulatory relationships encoded in each cell's full transcriptomic rank order.

Approach. Geneformer-V2-104M fine-tuned on HTAN SCLC T cells, then bidirectional *in silico* perturbation, with findings tested in intact tissue by quantitative spatial pathology.

WHAT IS AN *IN SILICO* PERTURBATION?

Geneformer reads a cell as an **ordered list of its genes**, most-expressed first. Deleting or inserting one gene in that list and re-reading the cell asks a counterfactual question: *if this gene changed, would the cell look more like a different disease state?*

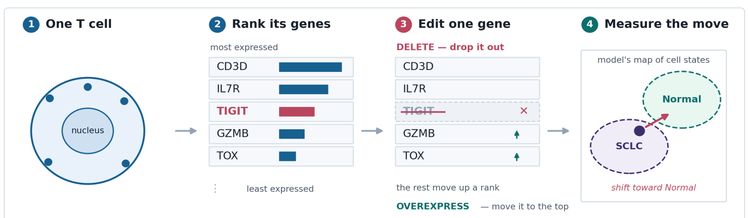


Fig. 1 — What an in silico perturbation does: rank a cell's genes, edit one of them, re-read the cell, measure how far it moved.

Nothing is edited in a laboratory here. The edit is made to the model's input and the result is the model's *predicted* change in cell state, averaged over thousands of real cells — a hypothesis generator, not a knockout experiment. A gene counts as a hit only when deleting and overexpressing it move cells in **opposite** directions, which is what the next panels report.

COHORT — HTAN MSK (CELLXGENE)

Disease	Train	Eval	Test	Donors
SCLC	7,037	2,330	2,424	**19**
LUAD	17,831	5,611	6,387	**22**
Normal	2,334	1,620	566	**4**

CD4 + CD8 T cells · donor-disjoint splits (seed 43) · cross-disease guard for paired tumour-normal donors · test split 8 donors.

Source: HTAN MSK, CELLxGENE "T cells" dataset 6fde3ad9 — **46,140 cells × 24,540 features** from **42 individuals** (45 donor×disease groups; median 390-1,172 (range 47-3,475) cells per donor). Donor leakage audit: **PASS**.

3 paired tumour-normal donors were confined to one split; the 91.9% test accuracy is therefore measured on individuals the model never saw.

B · MODEL, SCREEN & CANDIDATE HITS

CLASSIFIER PERFORMANCE — SCLC VS LUAD VS NORMAL

91.9%	0.903	9,377	8
Test accuracy	Macro F1	Held-out cells	Test donors

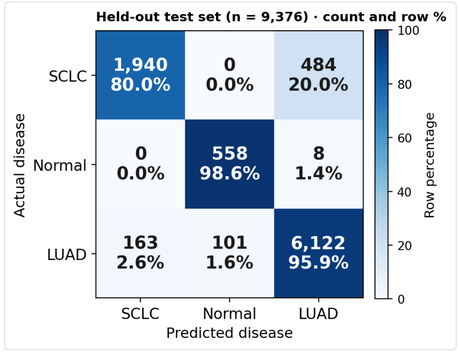


Fig. 2 — Held-out test confusion matrix.

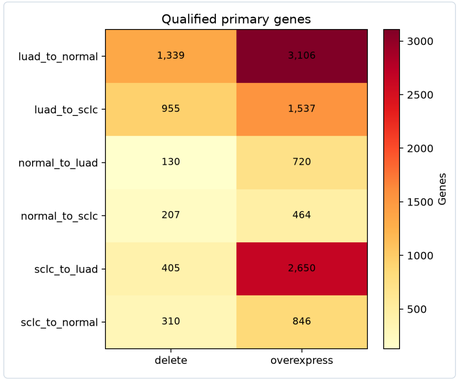


Fig. 3 — Qualified genes supporting the denoised programme.

Donor-held-out performance separates SCLC, Normal and LUAD sufficiently for the downstream perturbation screen. The antigen-presentation programme that led both the denoised screen and the spatial validation holds up alongside it: **no listed MHC gene is donor-inconsistent** in any transition, and unassessable entries reflect transitions with too few source donors rather than contradictory evidence.

DETECTION-AWARE APPLIED SCREEN — FOUR REPLICATED EDITS

50	4	–0.60	8
Screened genes	Replicated edits	Detection vs leffectl Spearman rho	No deletion result

Gene	Delete → Normal	Overexpress	Detected	Donors / tier
HAVCR2 (TIM-3)	**+0.0019**	-0.0060	202	3 / all donors agree
TIGIT	**+0.0018**	-0.0082	438	3 / all donors agree
CTLA-4	**+0.0014**	-0.0014	282	3 / all donors agree
IL7R (persistence)	**+0.0013**	-0.0054	1,131	3 / all donors agree

Values are mean cosine shifts toward Normal. A replicated edit is concordant at FDR < 0.05 in both arms, has the same sign in all 3 SCLC donors, and is detected in at least 100 source cells. These are reliable candidates, not the largest nominal effects.

C · SPATIAL PATHOLOGY — INDEPENDENT SCLC TISSUE COHORT

T-CELL DYSFUNCTION CO-LOCALISES WITH T-CELL ABUNDANCE IN TUMOUR TISSUE

0.161	4/5	15,632
Pooled Spearman rho [0.146, 0.176]	Specimens at p < 0.001	Tissue spots analysed

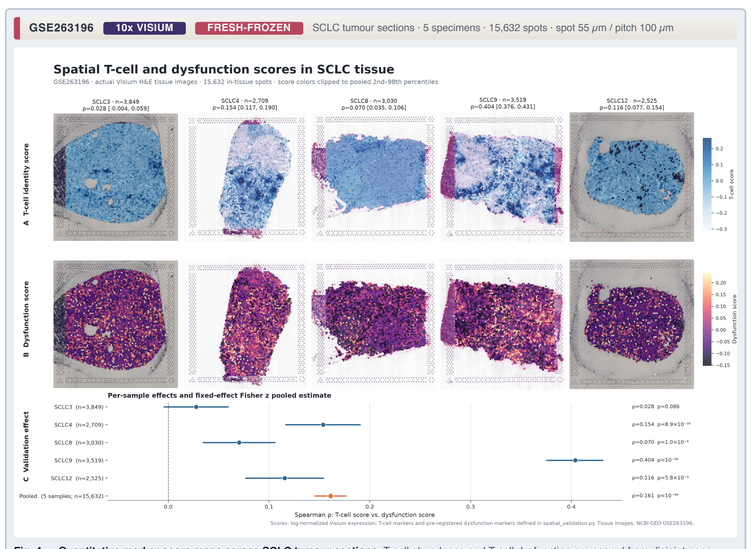


Fig. 4 — Quantitative marker-score maps across SCLC tumour sections. T-cell abundance and T-cell dysfunction are scored from disjoint gene panels at every spatially indexed spot, then correlated within each specimen.

Interpretation. In intact tumour tissue, regions with greater T-cell infiltration also carry higher co-inhibitory programme expression — the same association the single-cell perturbation screen implicates, recovered by an independent cohort and a different assay. Effect size varies substantially between specimens, which is consistent with real differences in tumour microenvironment composition rather than measurement noise.

Marker-score proxy rather than formal deconvolution; no Normal or LUAD sections, so SCLC-specificity of the association is untested; correlational design in archival tissue.

ONE DELETION AT A TIME, EVERY GENE, EVERY CELL — A PERTURB-SEQ SCREEN RUN *IN SILICO*

Fig. 5 — The screen in three acts: the Perturb-seq analogy and its scale (A); each deletion's measured effect on cell state, shown against the published STRING interaction map for the checkpoint genes (B); and the funnel from every gene-level test down to the four edits that replicate on every criterion (C).

WORKFLOW & VALIDATION HIERARCHY

HTAN T cells → Tokenise fine-tune → Disease classifier → Bidirectional perturbation → Spatial pathology

- **L1** Donor leakage audit — zero leakage across 42 donors; cross-disease guard for paired tumour-normal donors
- **L2** Held-out test — 91.9% accuracy, macro F1 0.903 (9,377 cells, 8 donors)
- **L3** Internal positive control — ASCL1 / NEUROD1 reproduce expected SCLC master regulator behaviour
- **L4** Bidirectional concordance — 123 concordant hits, 43 fully donor-consistent
- **L5** Spatial pathology — independent Visium cohort, pooled rho 0.161 [0.146, 0.176]

DIGITAL PATHOLOGY PIPELINE

Fresh-frozen tumour block → Cryosection onto array → Barcoded capture → Spot-level transcriptome → Quantitative score map

Each tissue section is resolved into thousands of spatially indexed transcriptomes. Two *disjoint* marker panels are scored per spot — T-cell abundance (CD3D/E/G, CD2, CD5, CD28, TRBC1/2, IL7R, CD8A/B, CD4) and T-cell dysfunction (PDCD1, CTLA4, HAVCR2, LAG3, TIGIT, TOX, LAYN) — so their correlation cannot be a self-correlation.

Platform 10x Visium | **Spot** 55 µm diameter | **Pitch** 100 µm centre-to-centre | **Specimens** 5 fresh-frozen SCLC

Spot geometry is the manufacturer platform specification, not a measurement from these sections. Marker scores are a proxy for composition, not deconvolved cell-type proportions.

DETECTION-AWARE CHECKPOINT SCREEN

The applied screen covers **50 checkpoint and T-cell engineering genes** in the SCLC → Normal comparison. Detection is joined on **(gene, comparison)**, because each source compartment has a different number of cells and a different expression profile. Eight genes have no deletion result: they were not detected in the SCLC source cells, so deletion is undefined, not zero.

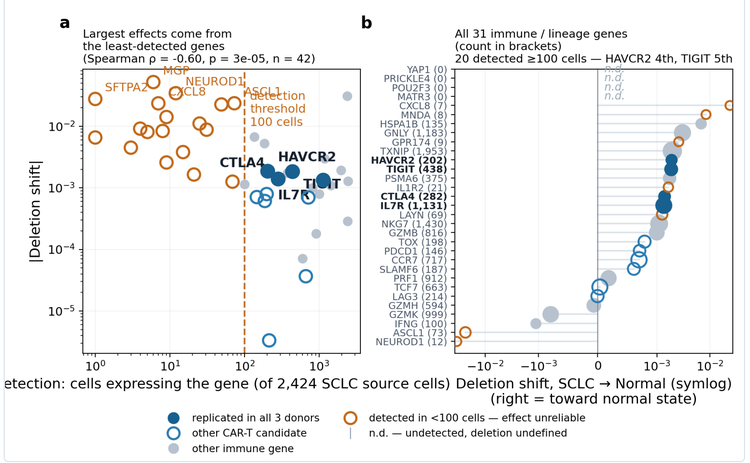


Fig. 6 — Detection versus absolute deletion effect across the full 50-gene panel.

Detection and nominal effect are strongly anti-correlated (Spearman ρ = **-0.60**, p = 2.7×10^{-5}, n = 42). The largest effects are therefore concentrated among sparsely detected genes and should not be read as reliable hits.

NETWORK CONTEXT & LITERATURE

The four replicated edits are not equally interpretable. **TIGIT** has the strongest external CAR-T support; **TIM-3** has direct knockout precedent in solid-tumour models. **CTLA-4** is a real signal with less CAR-T-specific evidence, while **IL7R** is a persistence target, not an exhaustion checkpoint.

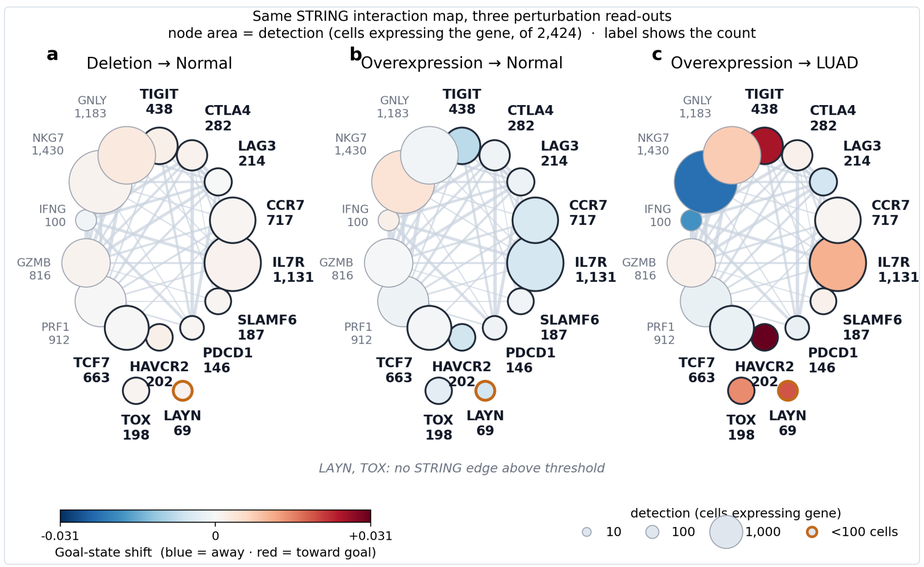


Fig. 7 — Shared interaction network under three perturbation readouts.

The network is a literature overlay: 54 filtered STRING edges among 16 context genes. TIM-3 is the strongest individual hit but has no supported non-text-mining edge to another checkpoint in this overlay, an explicit asymmetry rather than a plotting omission.

ICI / CAR-T CANDIDATES: EFFECT, REPLICATION, NETWORK

Three views of the same 11-gene ICI/CAR-T candidate set. Deletion shifts read as movement between disease states (a); restricted to the SCLC → normal transition, four knockouts replicate across all three SCLC donors (b); and the candidates are placed on the STRING interaction map, coloured by their SCLC → Normal deletion shift (c).

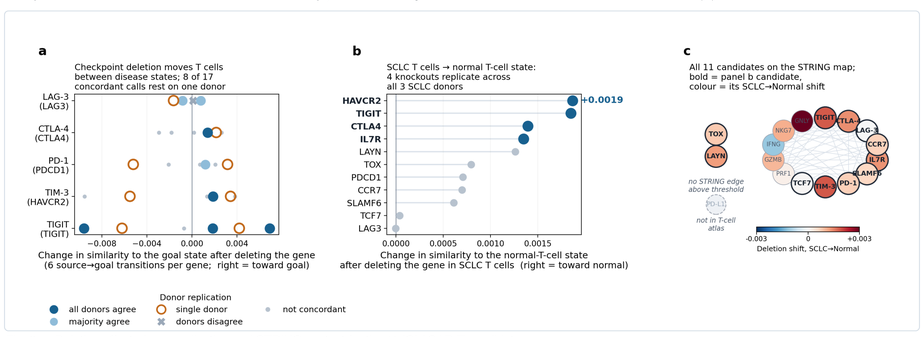


Fig. 8 — Deletion effect, donor replication and STRING context for the 11 ICI / CAR-T candidates.

8 of 17 concordant calls rest on a single donor, which is why the four-hit table above requires agreement across all three. PD-L1 is absent from the T-cell atlas, and TOX and LAYN carry no STRING edge above threshold.

LIMITATIONS

- CAR-T tumour antigens DLL3, SEZ6, NCAM1, CD276 and CEACAM5 are not perturbable in a T-cell dataset.
- Detection is source-state specific; 8 of 50 genes have no deletion result and 18 fall below 100 detected SCLC source cells.
- Perturbation shifts are model-derived interventions on a rank-value encoding, not experimental knockouts.

FUTURE DIRECTIONS

Immediate
- Raw CellRanger output from HTAN to enable a true CellBender run
- CD4 / CD8 subtype-stratified concordance

Spatial pathology
- cell2location deconvolution in place of marker scores
- Normal and LUAD sections as specificity controls

Mechanism
- Why MHC-I / antigen presentation dominates the denoised screen
- scGPT cross-model replication of the genome-scale screen

Experimental
- CRISPR knockout of TIGIT and HAVCR2 in SCLC co-culture
- Perturb-seq in primary T cells versus in silico predictions

VALIDATION COHORT — GSE263196 SPECIMENS & QC

Independent cohort: GSE263196, five fresh-frozen SCLC sections and 15,632 analysed spots. All 21 marker genes are present; no donor, cell or read is shared with the discovery atlas. This tests replication in intact tissue, not SCLC specificity.

D · TAKE-HOME & NEXT STEPS

CONCLUSIONS

- ✓ Geneformer-V2-104M fine-tuned on HTAN SCLC T cells reaches **91.9% accuracy, macro F1 0.903** on a donor-held-out test set.
- ✓ A detection-aware applied screen identifies four SCLC → Normal edits with full donor agreement, adequate detection, and FDR < 0.05 in both perturbation arms: TIM-3, TIGIT, CTLA-4, and IL7R.
- ✓ The detection-vs-effect-size confound is substantial (ρ = –0.60); sparse genes produce the largest nominal shifts, so effect magnitude alone is not a ranking criterion.
- ✓ TIGIT and TIM-3 have the strongest external CAR-T support. CTLA-4 is supported as a target class but has less CAR-T-specific evidence; IL7R is supported as a persistence gene, not a checkpoint.
- ✓ The model implies Normal < SCLC < LUAD on the checkpoint axis. This conflicts with the clinical tumour-level picture of SCLC as cold and ICI-resistant, so it remains an open question rather than a reconciled conclusion.

DENOISED IMMUNE/CANCER PROGRAMS

Immune/cancer genes prioritized on biological grounds are substantially more likely to reproduce across patients than the background of concordant hits. Because donor identity played no part in building those programs, this is independent support for the denoising step rather than a circular result.

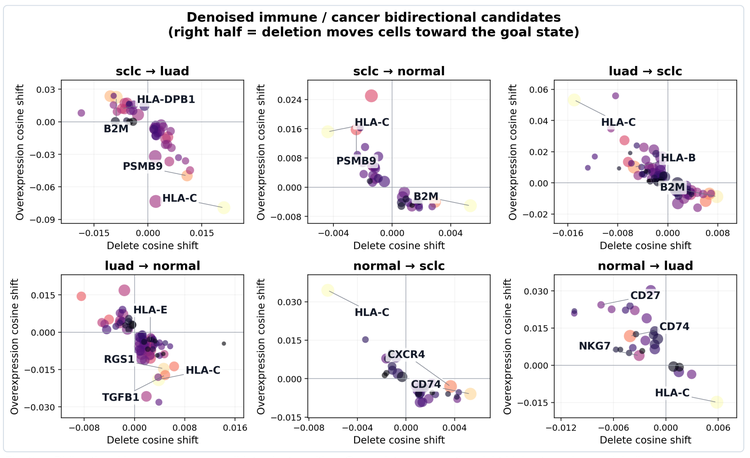


Fig. 9 — Bidirectional immune/cancer candidates by transition. One point per gene; the eight highest-scoring are labelled.

Colour — denoised balanced score, low to high
Size — detection, source cells the gene is seen in
x > 0 — deletion moves cells **toward** the goal state; y is the overexpression shift

The programme-level prioritization is supported by patient-level recurrence, not only by cell-weighted effect size.